Mihai-Cristian **Farcaș**

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EDUCATION

- **Babeș-Bolyai University** **Cluj-Napoca, Romania**
  BSc in Computer Science (English), Faculty of Mathematics and Computer Science *Oct 2023 - June 2026*
  - **Thesis**: Hybrid post-quantum key exchange for WireGuard, the first FIPS 203-compliant hybrid WireGuard implementation on a production Rust codebase.

EXPERIENCE

- **BluePlan** **Software Engineer (Freelance)**
  Atlas: RAG Platform *2026 - Present*
  - Built and shipped Atlas end to end: retrieval-augmented chat over technical documents with hybrid dense, sparse and image retrieval fused with RRF, reranking and verbatim citations pinned to the source PDF. Multi-tenant, multilingual and in daily production use. Python, FastAPI, Qdrant, React.
- **Baseline Studios** **Full-Stack Developer**
  Client Applications *2026*
  - Delivered full-stack features across multiple client applications (stratosphere.vip, pursuitofx.org, kabu.xyz) using TypeScript and React.
- **GeoDraw** **Software Developer**
  AI-Powered Design Canvas *2025*
  - Developed an AI-powered canvas for sketching and wireframing apps with React, TLDraw and LLM APIs, turning natural language into accurate real-time design output on the canvas.
- **Tvarita** **Software Development Intern**
  Website Redesign *April - June 2025*
  - Rebuilt and redesigned a server monitoring site for an optician clinic chain using React and Tailwind CSS, improving the user experience and significantly reducing load times.

PERSONAL PROJECTS

- **pq-boringtun** **2026**
  - Hybrid post-quantum key exchange for WireGuard: ML-KEM-768 alongside X25519 in the Noise IK handshake, built in Rust on Cloudflare's BoringTun and benchmarked on Apple M4 and Raspberry Pi 5.
- **Supernova** **2026**
  - Live-wallpaper engine for macOS in Swift 6: renders looping video, Shadertoy-style GLSL shaders transpiled to Metal and sandboxed web bundles behind desktop icons, with audio, cursor and motion reactivity, per-display assignment and smart playlists.
- **eBPF Network Analysis System** **2025**
  - Network traffic classification for QUIC protocols, inspecting encrypted flows at the kernel boundary using eBPF with C, Rust and Python.
- **nvim-search-and-replace** **2025**
  - Project-wide search and replace plugin for Neovim written in Lua: async streaming ripgrep search, live before/after preview, selective replacement and its own undo/redo stack.

AWARDS AND CERTIFICATIONS

- C2 Certificate of Proficiency in English (Cambridge English Assessment).
- Honorable mentions in multiple mathematical logic and computational thinking contests (e.g. BEBRAS).

TECHNICAL SKILLS

Programming Languages: Rust, Go, Python, Swift, C/C++, Java, TypeScript/JavaScript, Lua, Shell.

Technologies & Tools: React, Next.js, Astro, Tailwind CSS, FastAPI, PyTorch, LangChain, Qdrant, FAISS, PostgreSQL, eBPF, Metal, Git, Neovim, Cloudflare Workers.